

ITA0612-Machine Learning

PYTHON LIBRARY FILES IMPLEMENATATION

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1. Filtering Data

#1.Filtering Data

```
import pandas as pd

df = pd.read_csv("student_data.csv")

result = df[df["Marks"] > 80]

print(result)
```

	Name	Age	Department	Marks	Attendance
0	Arun	20	AI&DS	85.0	90.0
2	Charan	19	AI&DS	90.0	95.0
6	Gokul	23	ECE	88.0	NaN
8	Isha	19	AI&DS	92.0	96.0

2. Filtering Data with Multiple Conditions

#2.Filtering Data with Multiple Conditions

```
import pandas as pd

df = pd.read_csv("student_data.csv")

result = df[
    (df["Marks"] > 75) &
    (df["Attendance"] > 85)
]

print(result)
```

	Name	Age	Department	Marks	Attendance
0	Arun	20	AI&DS	85.0	90.0
2	Charan	19	AI&DS	90.0	95.0
5	Farhan	21	AI&DS	78.0	92.0
8	Isha	19	AI&DS	92.0	96.0

3. Sorting Data by Both Orders

#3. Sorting Data by Both Orders

#Ascending Order

```
import pandas as pd

df = pd.read_csv("student_data.csv")

result = df.sort_values(by="Marks", ascending=True)

print("Ascending Order \n",result,"\n")
```

#Descending Order

```
import pandas as pd

df = pd.read_csv("student_data.csv")

result = df.sort_values(by="Marks", ascending=False)

print("Descending Order \n",result)
```

Ascending Order

	Name	Age	Department	Marks	Attendance
3	Divya	22	ECE	65.0	80.0
9	Jaya	22	ECE	70.0	82.0
1	Bala	21	CSE	72.0	85.0
7	Hari	21	CSE	75.0	86.0
5	Farhan	21	AI&DS	78.0	92.0
0	Arun	20	AI&DS	85.0	90.0
6	Gokul	23	ECE	88.0	NaN
2	Charan	19	AI&DS	90.0	95.0
8	Isha	19	AI&DS	92.0	96.0
4	Esha	20	CSE	NaN	88.0

Descending Order

	Name	Age	Department	Marks	Attendance
8	Isha	19	AI&DS	92.0	96.0
2	Charan	19	AI&DS	90.0	95.0
6	Gokul	23	ECE	88.0	NaN
0	Arun	20	AI&DS	85.0	90.0
5	Farhan	21	AI&DS	78.0	92.0
7	Hari	21	CSE	75.0	86.0
1	Bala	21	CSE	72.0	85.0
9	Jaya	22	ECE	70.0	82.0
3	Divya	22	ECE	65.0	80.0
4	Esha	20	CSE	NaN	88.0

4. Handling Missing Values

```
#4. Handling Missing Values
#Detecting Missing Values
import pandas as pd

df = pd.read_csv("student_data.csv")

print("Detecting Missing Values\n",df.isnull(),"\n")

#Counting Missing Values
import pandas as pd

df = pd.read_csv("student_data.csv")

print("Counting Missing Values\n",df.isnull().sum(),"\n")

#Removing Missing Values
import pandas as pd

df = pd.read_csv("student_data.csv")

result = df.dropna()

print("Removing Missing Values\n", result)
```

Detecting Missing Values

	Name	Age	Department	Marks	Attendance
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	True	False
5	False	False	False	False	False
6	False	False	False	False	True
7	False	False	False	False	False
8	False	False	False	False	False
9	False	False	False	False	False

Counting Missing Values

Name	0
Age	0
Department	0
Marks	1
Attendance	1
dtype:	int64

Removing Missing Values

	Name	Age	Department	Marks	Attendance
0	Arun	20	AI&DS	85.0	90.0
1	Bala	21	CSE	72.0	85.0
2	Charan	19	AI&DS	90.0	95.0
3	Divya	22	ECE	65.0	80.0
5	Farhan	21	AI&DS	78.0	92.0
7	Hari	21	CSE	75.0	86.0
8	Isha	19	AI&DS	92.0	96.0
9	Jaya	22	ECE	70.0	82.0

5. Removing Duplicate Data

#5. Removing Duplicate Data

```
import pandas as pd
```

```
df = pd.read_csv("student_data.csv")
```

```
result = df.drop_duplicates()
```

```
print("Removing Duplicate Data\n", result)
```

Removing Duplicate Data

	Name	Age	Department	Marks	Attendance
0	Arun	20	AI&DS	85.0	90.0
1	Bala	21	CSE	72.0	85.0
2	Charan	19	AI&DS	90.0	95.0
3	Divya	22	ECE	65.0	80.0
4	Esha	20	CSE	NaN	88.0
5	Farhan	21	AI&DS	78.0	92.0
6	Gokul	23	ECE	88.0	NaN
7	Hari	21	CSE	75.0	86.0
8	Isha	19	AI&DS	92.0	96.0
9	Jaya	22	ECE	70.0	82.0

6. Aggregation

```
#6. Aggregation
import pandas as pd

df = pd.read_csv("student_data.csv")

print("Average Marks:")
print(df["Marks"].mean())

print("\nMaximum Marks:")
print(df["Marks"].max())

print("\nMinimum Marks:")
print(df["Marks"].min())

print("\nDepartment-wise Average Marks:")
print(df.groupby("Department")["Marks"].mean())
```

```
Average Marks:
79.44444444444444
```

```
Maximum Marks:
92.0
```

```
Minimum Marks:
65.0
```

```
Department-wise Average Marks:
Department
AI&DS      86.250000
CSE        73.500000
ECE        74.333333
Name: Marks, dtype: float64
```

7. Data Visualization

#7. Data Visualization

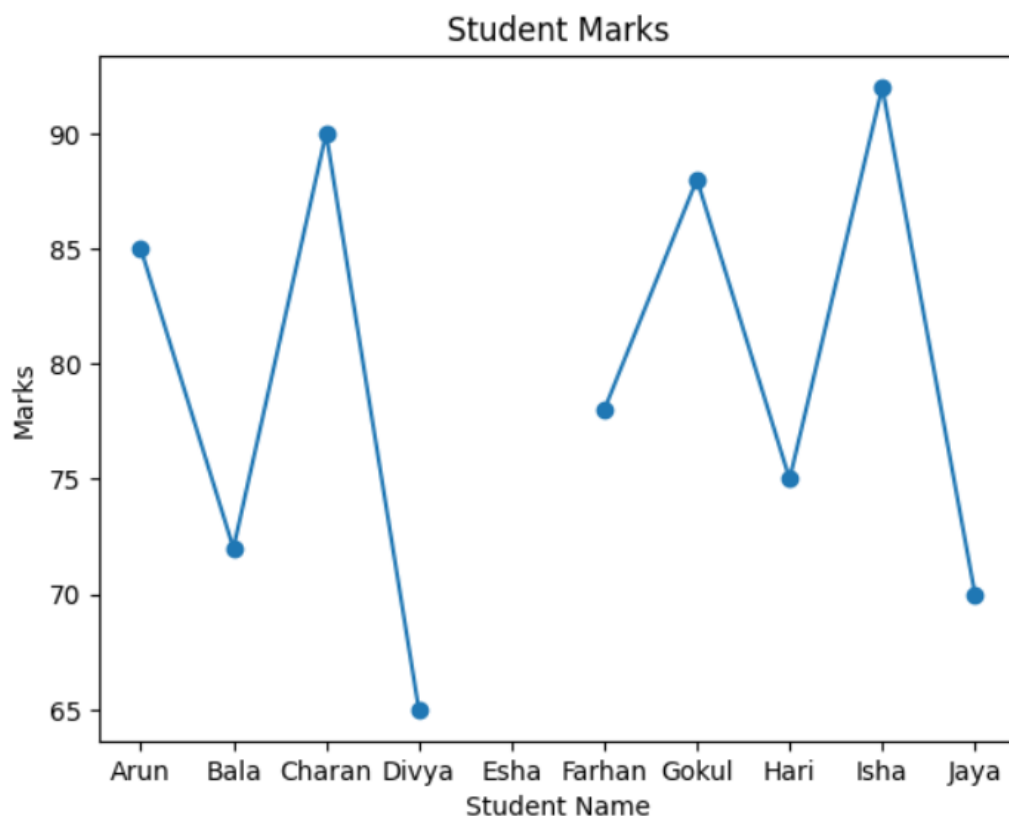
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("student_data.csv")

plt.plot(df["Name"], df["Marks"], marker="o")

plt.title("Student Marks")
plt.xlabel("Student Name")
plt.ylabel("Marks")

plt.show()
```



8. Line Plot

```
#8. Line Plot
import pandas as pd
import matplotlib.pyplot as plt

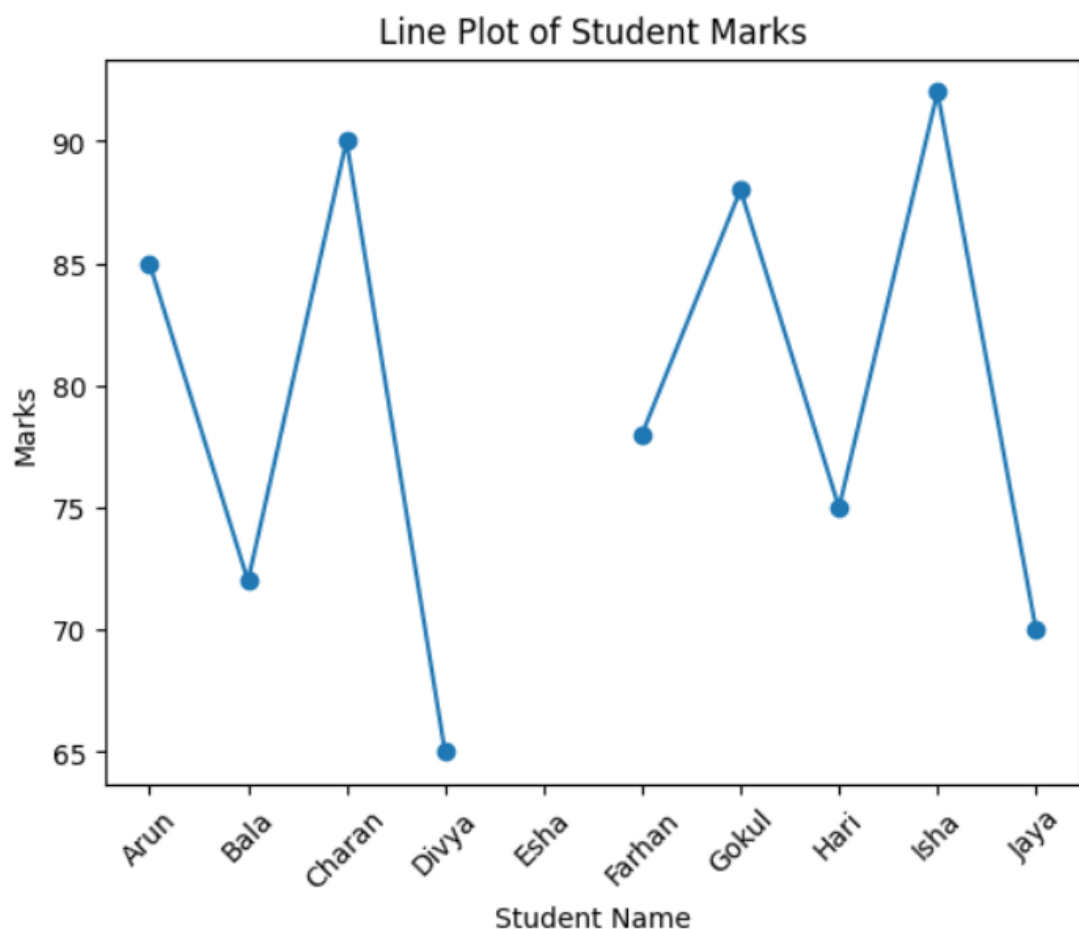
df = pd.read_csv("student_data.csv")

plt.plot(df["Name"], df["Marks"], marker="o")

plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.title("Line Plot of Student Marks")

plt.xticks(rotation=45)

plt.show()
```



9. Bar Chart

```
#9. Bar Chart
import pandas as pd
import matplotlib.pyplot as plt

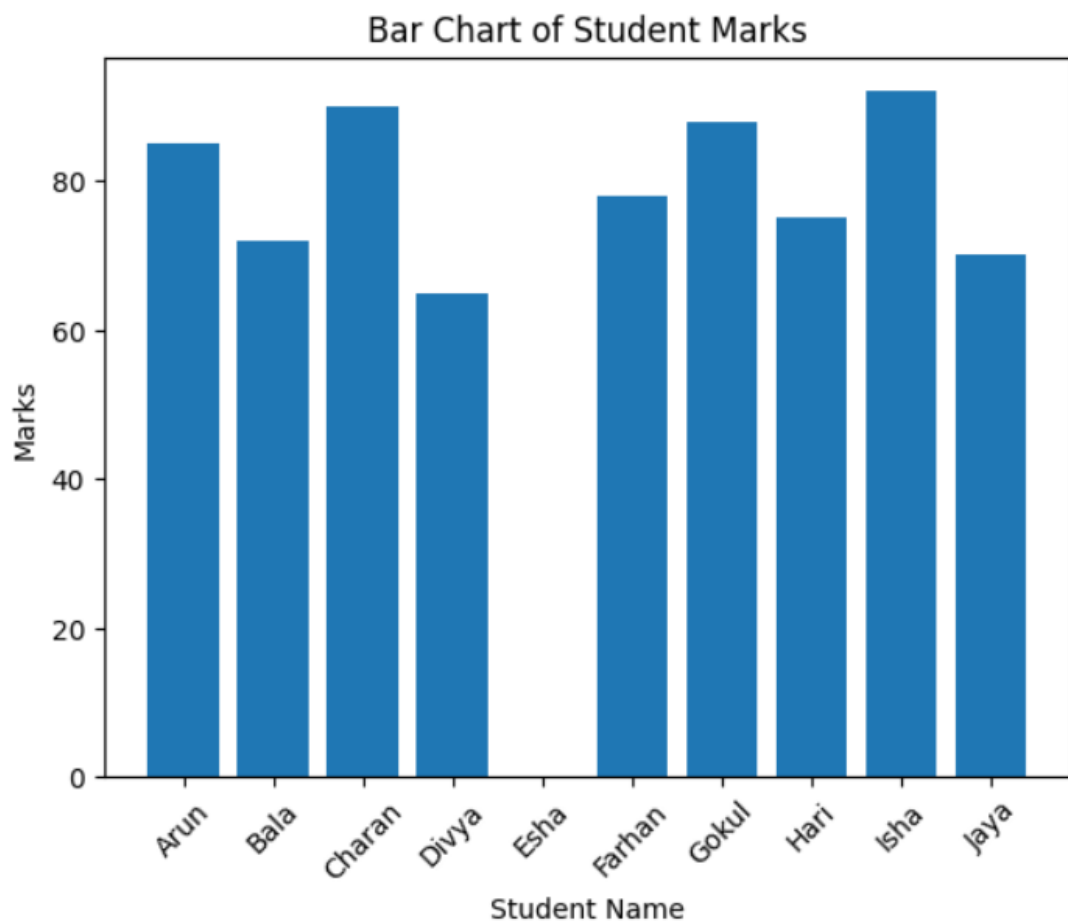
df = pd.read_csv("student_data.csv")

plt.bar(df["Name"], df["Marks"])

plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.title("Bar Chart of Student Marks")

plt.xticks(rotation=45)

plt.show()
```



10. Histogram

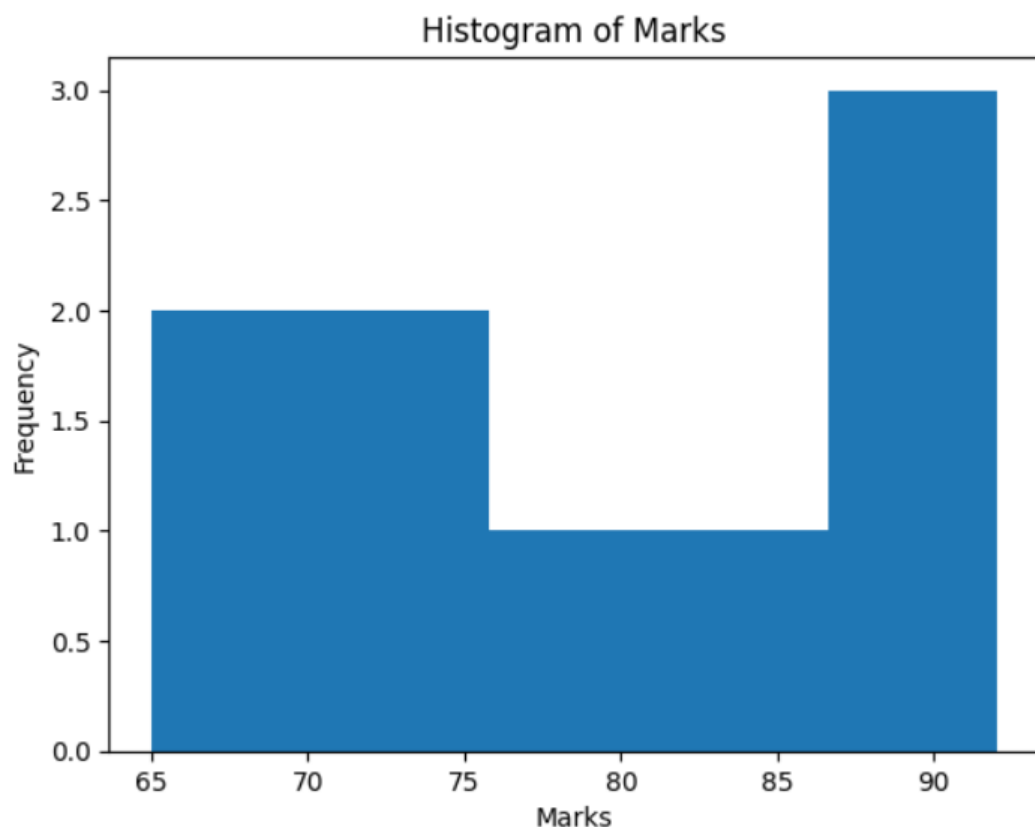
```
#10 Histogram
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("student_data.csv")

plt.hist(df["Marks"].dropna(), bins=5)

plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.title("Histogram of Marks")

plt.show()
```



11. Scatter plot

```
#11 Scatter Plot
import pandas as pd
import matplotlib.pyplot as plt

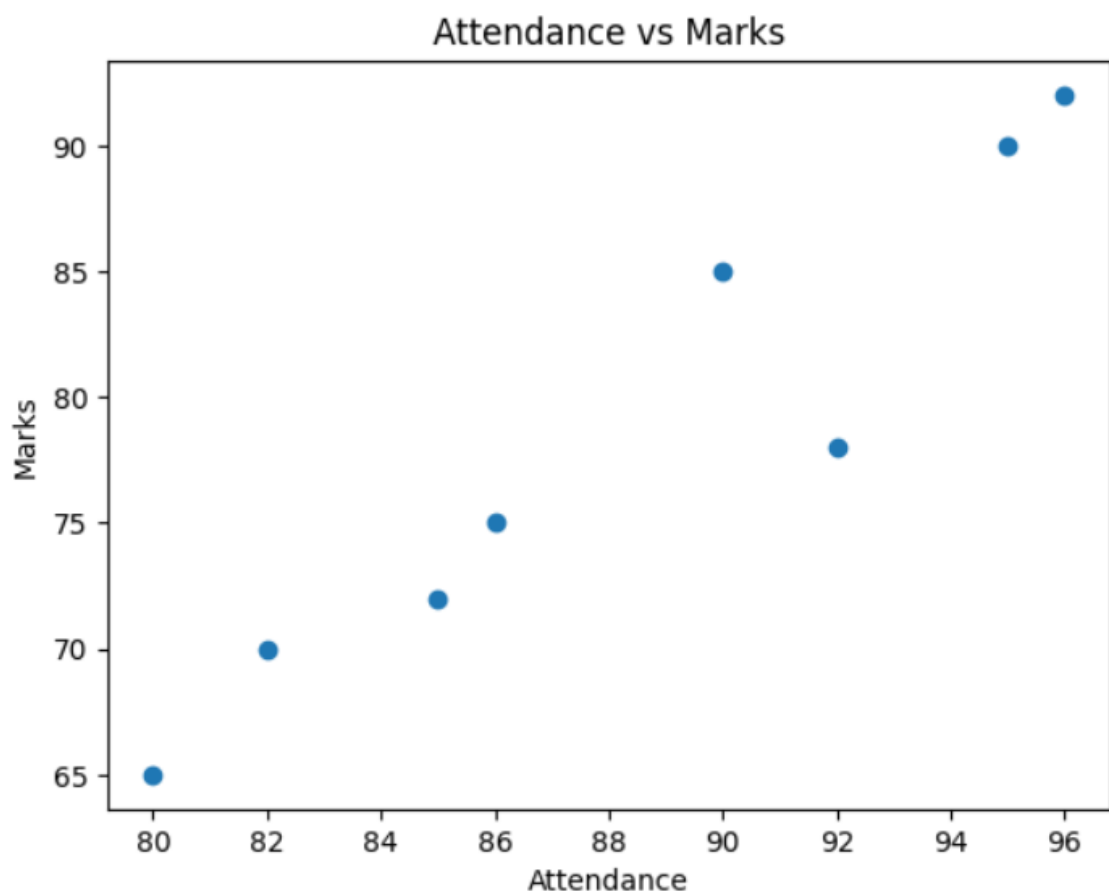
df = pd.read_csv("student_data.csv")

data = df.dropna(subset=["Marks", "Attendance"])

plt.scatter(data["Attendance"], data["Marks"])

plt.xlabel("Attendance")
plt.ylabel("Marks")
plt.title("Attendance vs Marks")

plt.show()
```



12. Pie Chart

```
#12. Pie Chart
import pandas as pd
import matplotlib.pyplot as plt

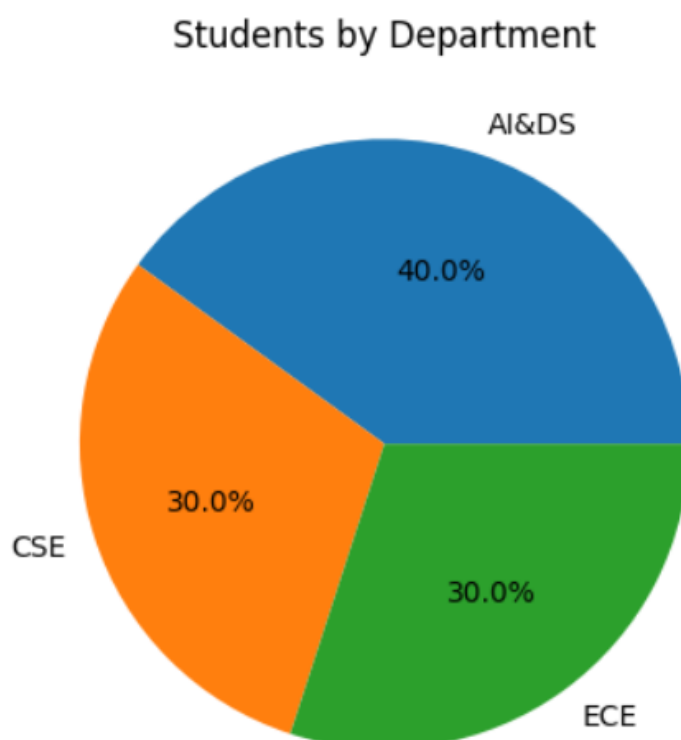
df = pd.read_csv("student_data.csv")

department_count = df["Department"].value_counts()

plt.pie(
    department_count,
    labels=department_count.index,
    autopct="%1.1f%%"
)

plt.title("Students by Department")

plt.show()
```



13. Plot Labels and Titles

```
#13. Plot Labels and Titles
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("student_data.csv")

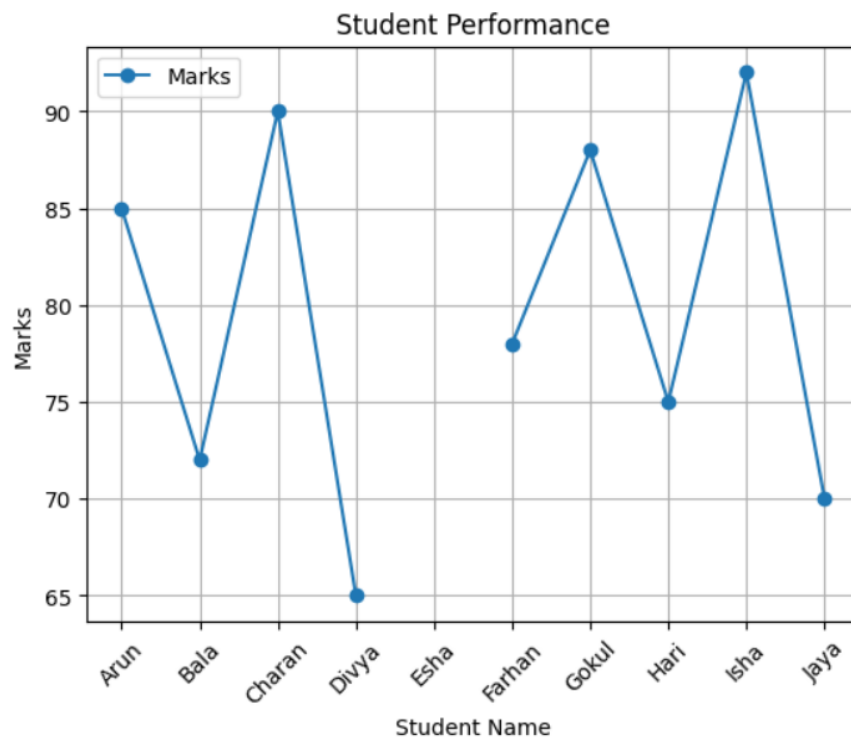
plt.plot(
    df["Name"],
    df["Marks"],
    marker="o",
    label="Marks"
)

plt.title("Student Performance")
plt.xlabel("Student Name")
plt.ylabel("Marks")

plt.legend()
plt.grid(True)

plt.xticks(rotation=45)

plt.show()
```



14. Sub Plots

```
#14.Subplots
import pandas as pd
import matplotlib.pyplot as plt

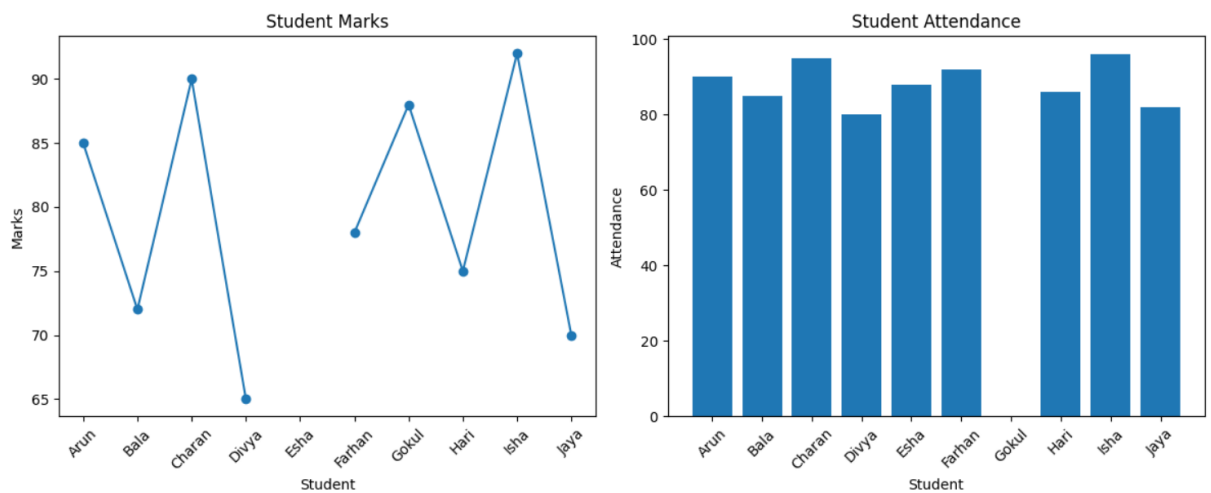
df = pd.read_csv("student_data.csv")

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Line Plot
axes[0].plot(df["Name"], df["Marks"], marker="o")
axes[0].set_title("Student Marks")
axes[0].set_xlabel("Student")
axes[0].set_ylabel("Marks")
axes[0].tick_params(axis="x", rotation=45)

# Bar Chart
axes[1].bar(df["Name"], df["Attendance"])
axes[1].set_title("Student Attendance")
axes[1].set_xlabel("Student")
axes[1].set_ylabel("Attendance")
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()
```



15. Reading and Writing to a CSV File

#15. Reading and Writing to a CSV File

```
import pandas as pd

df = pd.read_csv("student_data.csv")

print("Data read from CSV:")
print(df, "\n")

import pandas as pd

df = pd.read_csv("student_data.csv")

df.to_csv("student_data_output.csv", index=False)

print("CSV file written successfully.")
```

Data read from CSV:

	Name	Age	Department	Marks	Attendance
0	Arun	20	AI&DS	85.0	90.0
1	Bala	21	CSE	72.0	85.0
2	Charan	19	AI&DS	90.0	95.0
3	Divya	22	ECE	65.0	80.0
4	Esha	20	CSE	NaN	88.0
5	Farhan	21	AI&DS	78.0	92.0
6	Gokul	23	ECE	88.0	NaN
7	Hari	21	CSE	75.0	86.0
8	Isha	19	AI&DS	92.0	96.0
9	Jaya	22	ECE	70.0	82.0

CSV file written successfully.